

GenAI Assignment 4

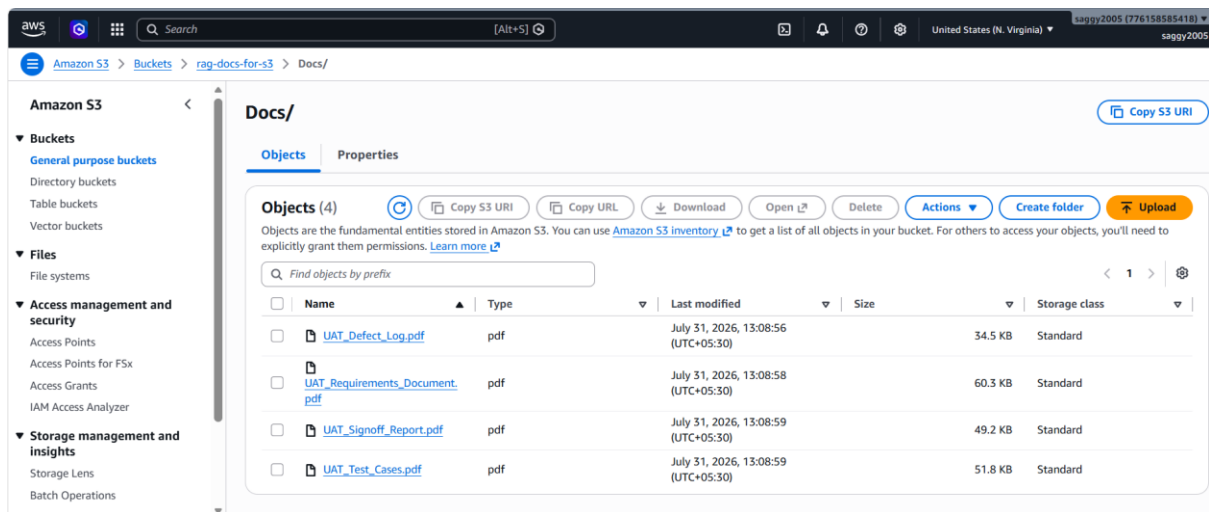
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The entire Project is done in us-east-1 server

Step 1: Model Comparison

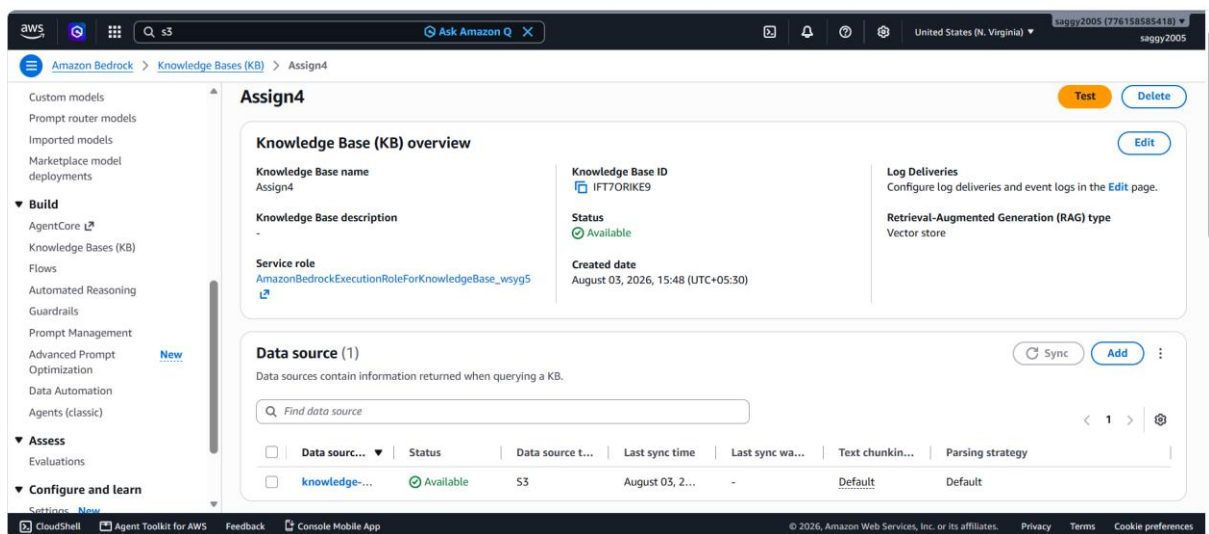
(A separate word file is attached in the Assignment 4 file repository)

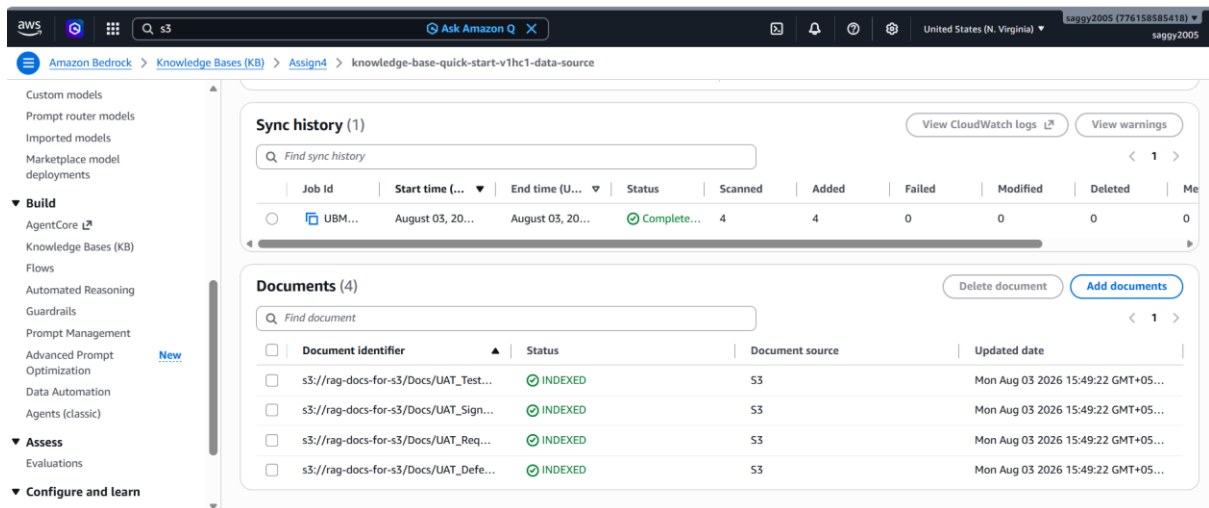
Step 2: Creation of RAG Bot



A bucket is created with all the necessary folders uploaded.

Step 3: Create Knowledge Base

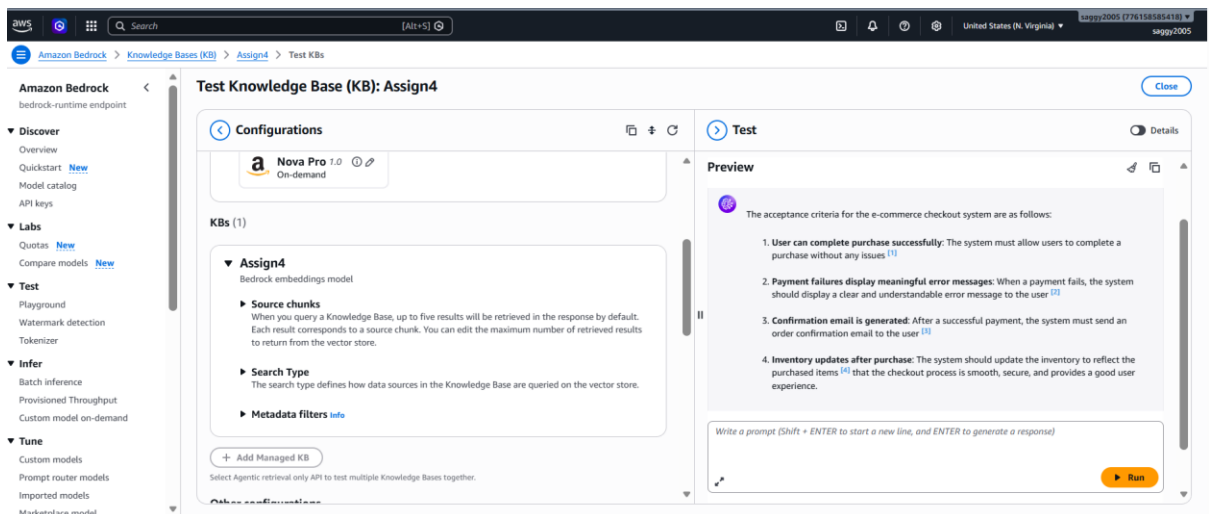




Unstructured Vector Store Knowledge Base has been created with data source from S3 bucket.

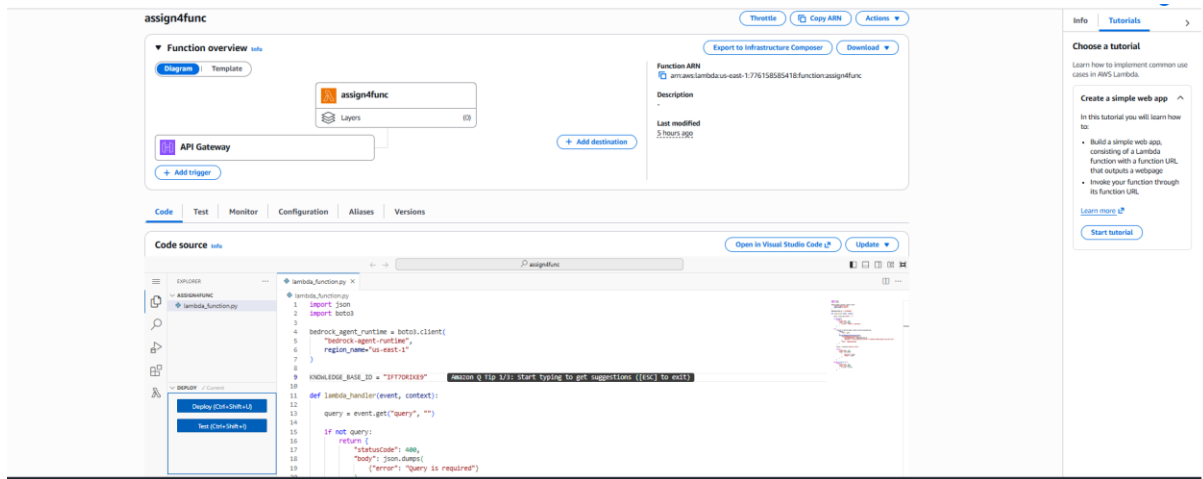
Step 4: Test The knowledge base

Amazon Nova Pro Model, which was inferred as the best model during comparison, has been selected for this testing.

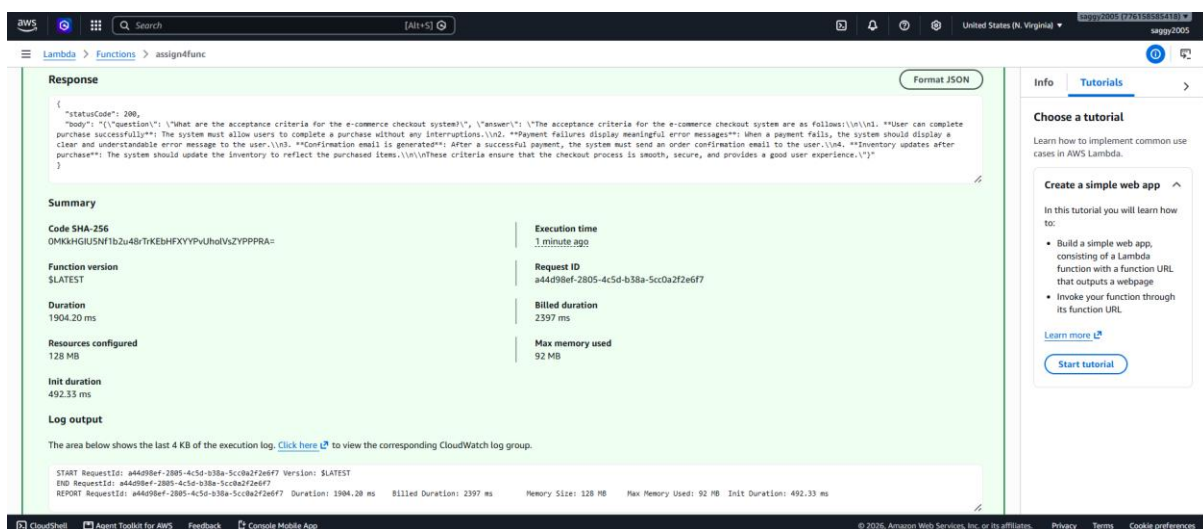
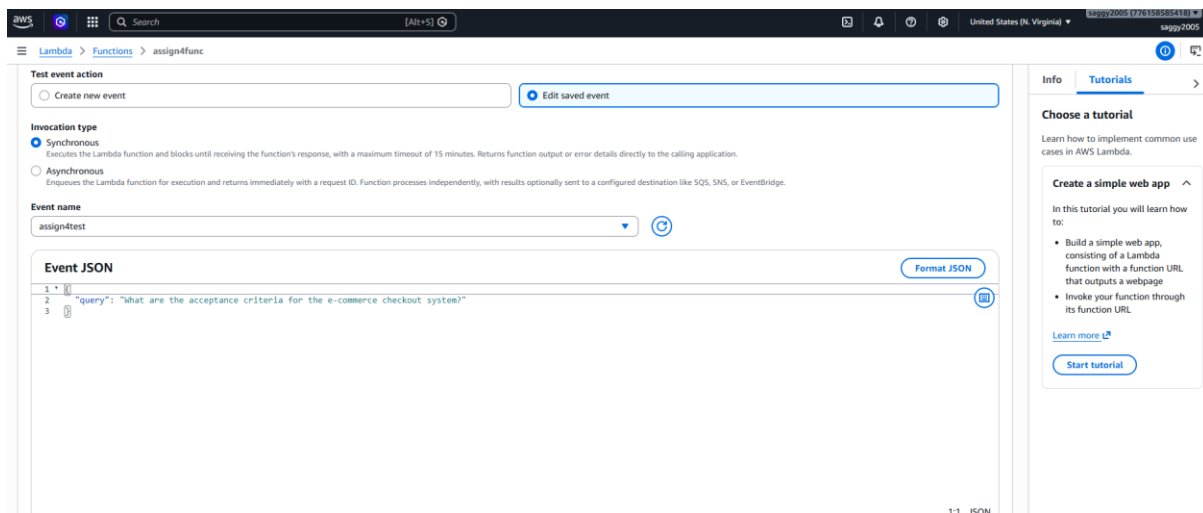


When we put the prompt, it successfully fetches the required data from the S3 bucket

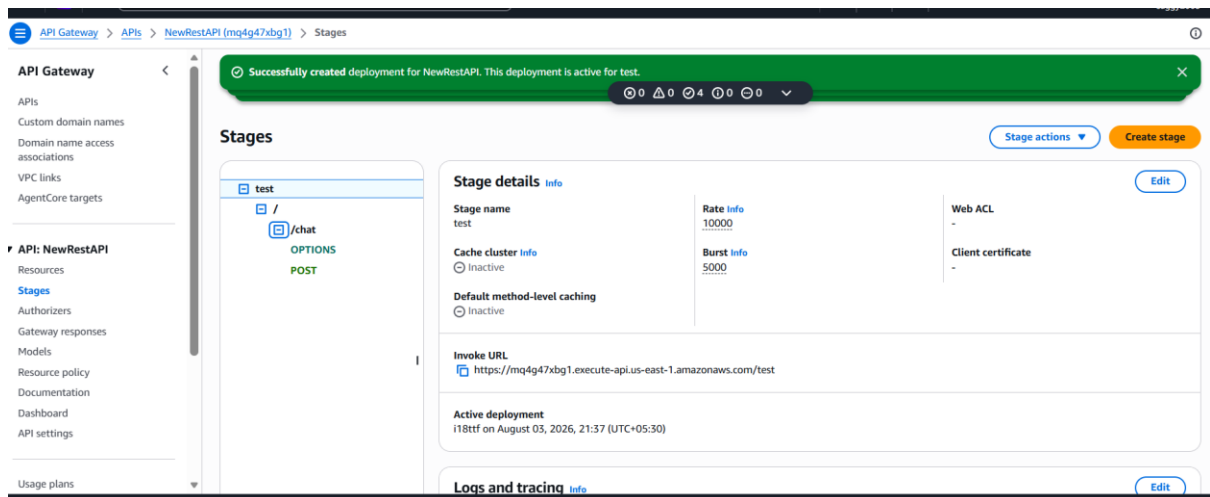
Step 5: Create Lambda function



Step 6: Test Lambda Function

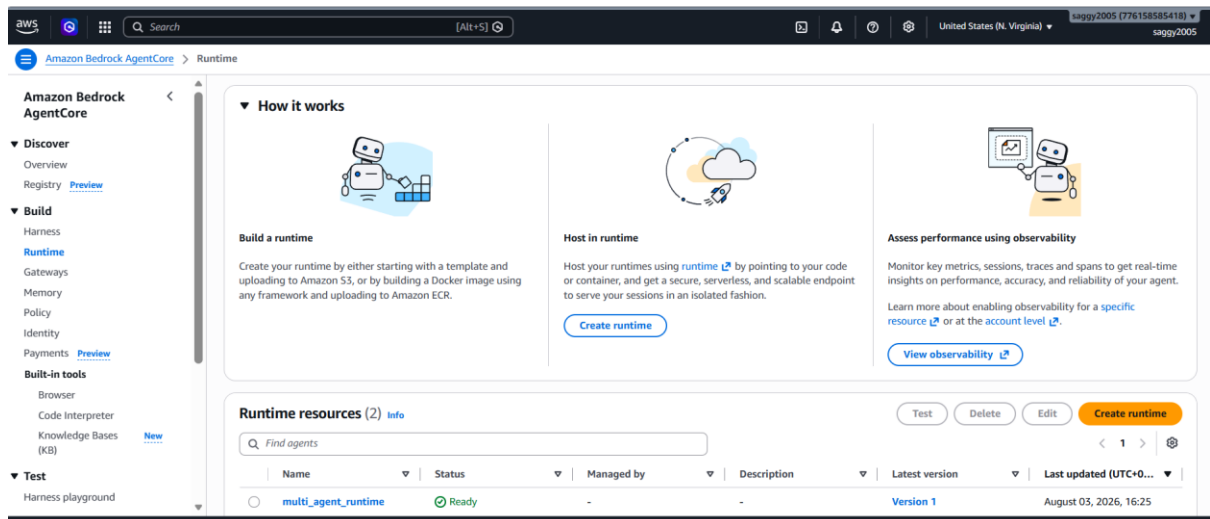


Step 7: Create API



REST API with POST

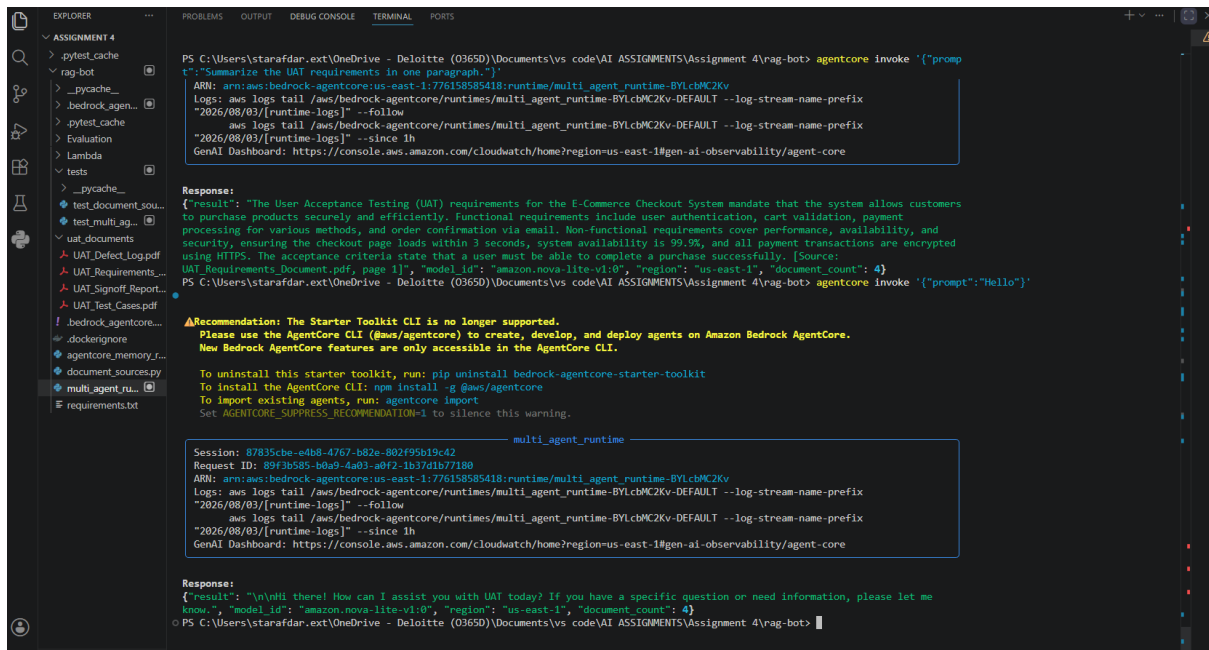
Step 8: Agent Creation



Agent has been created through VS Code powershell terminal

Step 9: Agentcore Invoke

To check if the agent is properly working and responding to queries and prompts



```
PS C:\Users\starafdar.ext\OneDrive - Deloitte (0365D)\Documents\vs code\AI ASSIGNMENTS\Assignment 4\rag-bot> agentcore invoke '{"prompt": "Summarize the UAT requirements in one paragraph."}'
ARN: arn:aws:bedrock:agentcore:us-east-1:776158585418:runtime/multi_agent_runtime-BYLCbMC2Kv
Logs: aws logs tail /aws/bedrock-agentcore/runtimes/multi_agent_runtime-BYLCbMC2Kv-DEFAULT --log-stream-name-prefix "2026/08/03/[runtime-logs]" --follow
aws logs tail /aws/bedrock-agentcore/runtimes/multi_agent_runtime-BYLCbMC2Kv-DEFAULT --log-stream-name-prefix "2026/08/03/[runtime-logs]" --since 1h
GenAI Dashboard: https://console.aws.amazon.com/cloudwatch/home?region=us-east-1#gen-ai-observability/agent-core

Response:
{"result": "The User Acceptance Testing (UAT) requirements for the E-Commerce Checkout System mandate that the system allows customers to purchase products securely and efficiently. Functional requirements include user authentication, cart validation, payment processing for various methods, and order confirmation via email. Non-functional requirements cover performance, availability, and security, ensuring the checkout page loads within 3 seconds, system availability is 99.9%, and all payment transactions are encrypted using HTTPS. The acceptance criteria state that a user must be able to complete a purchase successfully. [Source: UAT_Requirements_Document.pdf, page 1]", "model_id": "amazon.nova-lite-v1.0", "region": "us-east-1", "document_count": 4}
PS C:\Users\starafdar.ext\OneDrive - Deloitte (0365D)\Documents\vs code\AI ASSIGNMENTS\Assignment 4\rag-bot> agentcore invoke '{"prompt": "Hello"}'

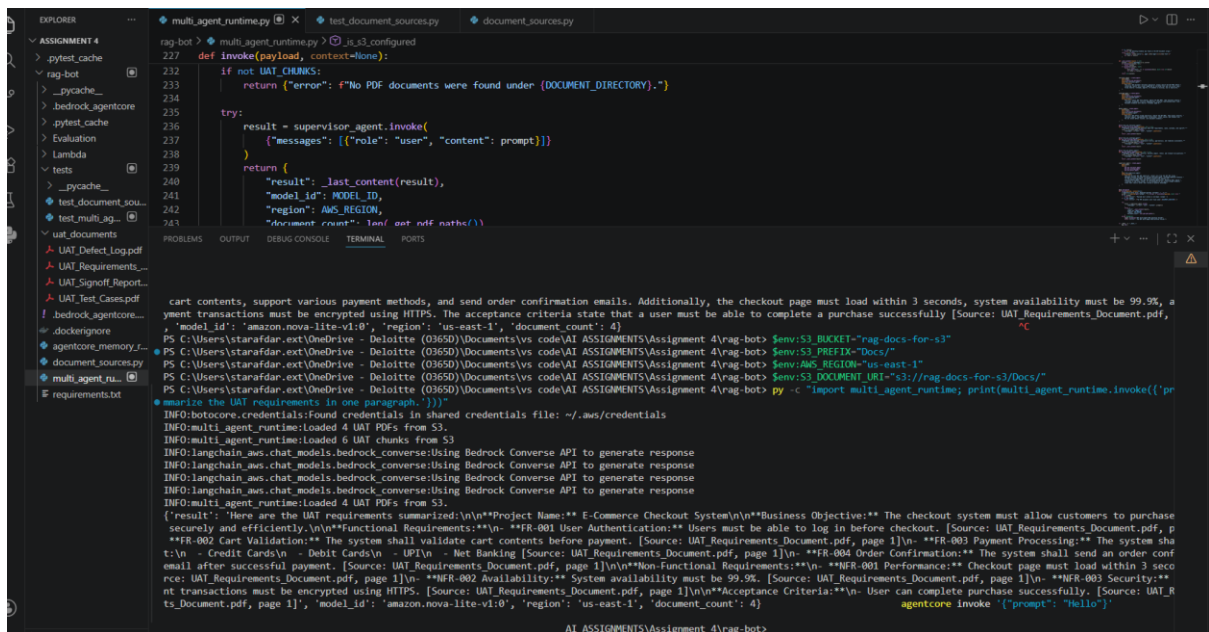
Recommendation: The Starter Toolkit CLI is no longer supported.
Please use the AgentCore CLI (#aws/agentcore) to create, develop, and deploy agents on Amazon Bedrock AgentCore.
New Bedrock AgentCore features are only accessible in the AgentCore CLI.

To uninstall this starter toolkit, run: pip uninstall bedrock-agentcore-starter-toolkit
To install the AgentCore CLI: npm install -g @aws/agentcore
To import existing agents, run: agentcore import
Set AGENTCORE_SUPPRESS_RECOMMENDATION=1 to silence this warning.

Session: 87835cbe-edb8-4767-b82e-802f95b19c42
Request ID: 89f3b585-b0a9-4a03-adf2-1b37d1b77180
ARN: arn:aws:bedrock-agentcore:us-east-1:776158585418:runtime/multi_agent_runtime-BYLCbMC2Kv
Logs: aws logs tail /aws/bedrock-agentcore/runtimes/multi_agent_runtime-BYLCbMC2Kv-DEFAULT --log-stream-name-prefix "2026/08/03/[runtime-logs]" --follow
aws logs tail /aws/bedrock-agentcore/runtimes/multi_agent_runtime-BYLCbMC2Kv-DEFAULT --log-stream-name-prefix "2026/08/03/[runtime-logs]" --since 1h
GenAI Dashboard: https://console.aws.amazon.com/cloudwatch/home?region=us-east-1#gen-ai-observability/agent-core

Response:
{"result": "\nHi there! How can I assist you with UAT today? If you have a specific question or need information, please let me know.", "model_id": "amazon.nova-lite-v1.0", "region": "us-east-1", "document count": 4}
PS C:\Users\starafdar.ext\OneDrive - Deloitte (0365D)\Documents\vs code\AI ASSIGNMENTS\Assignment 4\rag-bot>
```

Step 10: Output



```
multi_agent_runtime.py X test_document_sources.py document_sources.py
rag-bot > multi_agent_runtime.py > @_is_s3_configured
227 def invoke(payload, context=None):
228     if not UAT_CHUNKS:
229         return ("error": f"No PDF documents were found under {DOCUMENT_DIRECTORY}.")
230
231     try:
232         result = supervisor_agent.invoke(
233             {"messages": [{"role": "user", "content": prompt}]}
234         )
235         return {
236             "result": _last_content(result),
237             "model_id": MODEL_ID,
238             "region": AWS_REGION,
239             "document count": len(art_rdf_naths())
240         }
241     except Exception as e:
242         return {"error": str(e)}
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Output successfully prints by reading documents via S3 bucket.